

CarbonLens RAG Knowledge Context

Carbon Credit Quality, Integrity and Buyer Evaluation

Purpose: concise, retrieval-friendly context derived from three selected public sources. Use it to explain quality, risk, registries, methodologies, verification, permanence and price context. Do not treat this document as an official rating methodology.

1. Core concepts

- **Carbon credit:** For this platform, one carbon credit is generally represented as one tonne of carbon dioxide equivalent (1 tCO₂e).
- **Project:** A real-world activity that seeks to reduce, avoid or remove greenhouse-gas emissions and may generate credits under an approved methodology.
- **Registry / carbon-crediting program:** A system that can identify, record and track mitigation activities and issued credits.
- **Methodology:** The defined approach used to determine how emissions reductions or removals are measured and credited.
- **Vintage:** The year associated with the emissions reduction or removal represented by a credit. It is useful context, not a complete measure of quality.
- **Credits issued:** The number of credits issued for a project. This does not necessarily mean those credits are currently available for purchase.

2. ICVCM Core Carbon Principles

The Integrity Council for the Voluntary Carbon Market presents ten science-based Core Carbon Principles as a benchmark for identifying high-quality credits and real, verifiable climate impact. They cover governance, emissions impact and sustainable impact.

- **Effective governance:** Transparency, accountability, continuous improvement and overall credit quality.
- **Tracking:** Credits and mitigation activities should be uniquely identified, recorded and tracked.
- **Transparency:** Information about credited activities should be comprehensive, public and accessible for scrutiny.
- **Independent validation and verification:** Programs should have robust independent third-party validation and verification requirements.
- **Additionality:** The credited reduction or removal should not have occurred without the incentive created by carbon-credit revenue.
- **Permanence:** Climate benefits should last, or reversal risks should be addressed and compensated for.
- **Robust quantification:** Reductions and removals should be conservatively and scientifically quantified.
- **No double counting:** Mitigation outcomes should not be double issued, double claimed or double used.
- **Sustainable development and safeguards:** Activities should follow strong social and environmental safeguards.
- **Contribution toward net zero transition:** Activities should avoid locking in emissions or practices incompatible with net zero.

CarbonLens interpretation

These principles can inform explanations and risk-oriented reasoning. Registry transparency and independent verification can support confidence; methodology and quantification affect confidence in measurement; permanence is relevant to reversal risk. Do not claim a project is CCP-approved without reliable supporting data.

3. Calyx Global: CCP labels and remaining risk

- The CCP framework establishes a minimum benchmark for carbon-credit quality.
- A carbon-credit category combines project type and greenhouse-gas methodology.
- CCP labeling involves assessment at both the carbon-crediting program and category levels.
- A CCP label can establish a minimum threshold, but it does not mean every approved credit has identical quality or zero risk.
- Risk can still vary within approved categories.
- For CarbonLens, project type alone should not automatically determine a perfect score; methodology, verification, registry information and other available evidence should matter.

4. Berkeley buyer guidance: methodology and evidence matter

The Berkeley Carbon Trading Project's cookstove guidance illustrates a broader lesson: evaluation should not stop at the project label. Methodology, monitoring, assumptions and project documents can materially affect confidence in claimed climate benefits.

- **Direct measurement and monitoring:** Where suitable, direct or continuous measurement can provide stronger evidence than assumptions or indirect estimates.
- **Conservative calculation:** Methodologies should avoid overstating emissions reductions; uncertainty and behavioural effects may require conservative adjustments.
- **Project documents matter:** Project descriptions, validation reports, monitoring reports and verification reports provide different types of evidence.
- **Recent operational evidence matters:** Monitoring and verification can help show what actually happened, not only what was planned.
- **Registry records are useful:** Major registries can provide searchable project information and supporting documentation.

5. Suggested rules for the CarbonLens AI assistant

- Never say a higher CarbonLens score proves official certification as high integrity.
- Treat the CarbonLens score as a platform-generated heuristic based on available data and defined rules.
- Explain positive factors and risks rather than simply declaring a project good or bad.
- Never infer current availability for purchase from `credits_issued` or `volume_issued`.
- When discussing price, state that the platform price is an estimated reference price per carbon credit / per tCO₂e, not a guaranteed transaction price.
- Do not treat registry name alone as sufficient proof of quality.
- Do not treat project type alone as sufficient proof of quality.
- If relevant information is missing, say that the platform cannot assess that factor from available data.

- Avoid inventing official approvals, CCP labels, verification results, project documents or market prices.

6. Retrieval-friendly example answers

What does permanence mean?

Permanence refers to whether a claimed emissions reduction or carbon removal will last over time. Some activities face reversal risks, so high-integrity approaches should address those risks and, where appropriate, compensate for reversals.

Does a recognized registry guarantee a perfect credit?

No. Registry governance can be a positive factor, but quality can still depend on methodology, project category, quantification, monitoring, verification and project-specific risks.

Are credits issued the same as credits available to buy?

Not necessarily. Credits issued describe how many credits were issued; current availability requires separate information about ownership, retirement and market offers.

What does the displayed price mean?

The CarbonLens price is an estimated reference price per carbon credit, equivalent to a price per tonne of CO₂ equivalent. It is not a guaranteed purchase or transaction price.

7. Source notes

This document summarizes and adapts the selected sources for retrieval use. The AI should prefer project-specific data supplied to it over broad generalisations.

ICVCM — The Core Carbon Principles: <https://icvcm.org/core-carbon-principles/>

Calyx Global — The ICVCM Core Carbon Principle Labels:
<https://calyxglobal.com/research-hub/commentary/the-icvcm-core-carbon-principle-labels/>

Berkeley Carbon Trading Project — Guidance for Cookstove Offset Buyers:
<https://gspp.berkeley.edu/berkeley-carbon-trading-project/cookstoves/guidance-for-cookstove-offset-buyers>